

Notes from Jeremy's IT Lab (Days 11-15)

Routing Fundamentals

Routing is the process that routers use to determine the path that IP packets should take over a network to reach their destination. Routers store routes to all of their known destinations in a routing table. When routers receive packets, they look in the routing table to find the best route to forward that packet.

Dynamic routing:

Routers use dynamic routing protocols (e.g., OSPF) to share routing information with each other automatically and build their routing tables.

Static routing:

A network engineer/admin manually configures routes on the router.

A route tells the router what to do with a packet based on its destination IP address:

- If the destination is on a remote network, send the packet to the appropriate next hop. The next hop is the next router in the path toward the destination.
- If the destination is on a directly connected network, send the packet directly to the destination.
- If the destination IP address belongs to the router itself, the router receives and processes the packet instead of forwarding it.

To view the routing table, run:

- show ip route

```
Router(config-if)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.1.0/25 is directly connected, GigabitEthernet0/0
L       192.168.1.126/32 is directly connected, GigabitEthernet0/0
C       192.168.1.128/25 is directly connected, GigabitEthernet0/1
C       192.168.1.254/32 is directly connected, GigabitEthernet0/1
```

The Codes legend in the output of 'show ip route' lists the different protocols the router can use to learn routes.

L - local:

A route to the actual IP address configured on the interface (with a /32 net mask).

C - connected:

A route to the network the interface is connected to (with the actual net mask configured on the interface).

When you configure an IP address on an interface and enable it with 'no shutdown', 2 routes (per interface) will automatically be added to the routing table.

- a connected route
- a local route

A **connected route** represents the network that a router interface is directly connected to. It tells the router that hosts within that network can be reached through that interface.

- For example, if an interface is configured with 192.168.1.1/24, the router creates a connected route for 192.168.1.0/24.

A **local route** represents the exact IP address configured on a router interface.

- Local routes use a /32 prefix, meaning all 32 bits of the IP address are fixed and the route matches only that single address. This allows the router to recognize packets addressed specifically to one of its own interface IP addresses and process them locally instead of forwarding them.

Route Matching and Route Selection

A route matches a packet if the packet's destination IP address falls within the network specified by that route.

If multiple routes match the destination, the router chooses the **most specific** matching route.

The most specific matching route is the route with the **longest prefix length**.

For example, a packet destined for 192.168.1.1 may match both:

Connected route: 192.168.1.0/24

Local route: 192.168.1.1/32

- Because /32 is more specific than /24, the router selects the local route and recognizes that the packet is destined for itself rather than forwarding it onto the connected network.

Most specific matching route = matching route with the longest prefix length.

End hosts can send packets directly to a destination in their own connected network; however, to send packets to destinations outside of their local network they must send the packets to their **default gateway**.

The default gateway configuration is also called a **default route**.

The default route is a route to 0.0.0.0/0 - includes all addresses from 0.0.0.0 to 255.255.255.255.

Important:

The default route says if there aren't any more specific matching routes for this packet, send it to me instead of dropping it.

- 'Gateway of last resort is not set' means no default route has been configured yet.

Configuring a default route follows the same syntax as configuring a static route, the only exception is that the (ip-address net-mask) should be set to 0.0.0.0.

Example: ip route 0.0.0.0 0.0.0.0 *next-hop*

Static routes are manually configured entries in a router's routing table that specifies the next hop IP address or exit interface used to reach a destination network.

To configure a static route using next hop, run:

- ip route *ip-address netmask next-hop*

This tells the router where to send it based on IP, but the router has to consult its routing table to identify the interface needed to reach said IP.

The syntax is this: To reach xxx.xxx.xxx.xxx with netmask xxx.xxx.xxx.xxx, send to next-hop (this IP address)

To configure a static route using an exit interface:

- ip route *ip-address netmask exit-interface*

This is telling the router where to go, but only providing what 'door' it needs to send it out of, not necessarily telling it where it's going, just where it should send off from.

To configure a fully specified static route using both the exit interface and next hop:

- ip route *ip-address netmask exit-interface next-hop*

The 'S' in the routing table stands for static route - the 1/0 in the square brackets represent the **administrative distance/metric**.

Static routes in which you specify only the exit-interface rely on a feature called the proxy arp function. Per Jeremy, this is beyond the scope of the CCNA exam.

SUBNETTING

Subnetting is the process of dividing one IP network into multiple smaller networks by extending the network portion of the address into bits that were previously available for hosts.

In modern IPv4, this is controlled by the subnet mask or CIDR prefix length. The prefix tells you how many of the 32 bits belong to the network portion.

Example:

192.168.1.0/24

- A /24 means: 24 network bits, 8 host bits

If that /24 is divided into /26 subnets:

/24 -> /26

Two additional bits have been taken from the host portion and used as subnet bits.

- Network bits: 26
- Host bits: 6

Those 2 borrowed bits create:

- $2^2 = 4$ subnets

The remaining 6 host bits allow:

- $2^6 = 64$ total addresses per subnet

- $64 - 2 = 62$ usable host addresses

The subtraction accounts for the network address and broadcast address of each subnet.

A 192.168.1.0/24 network divided into /26 subnets becomes:

- 192.168.1.0/26
- 192.168.1.64/26
- 192.168.1.128/26
- 192.168.1.192/26

Each subnet is its own independent Layer 3 network with its own:

Network address

Usable host range

Broadcast address

Example:

192.168.1.64/26

Network address: 192.168.1.64

First usable host: 192.168.1.65

Last usable host: 192.168.1.126

Broadcast address: 192.168.1.127

SUBNETTING CALCULATIONS

Once the prefix is known, most subnetting problems come down to two main questions:

1. How many addresses are in each subnet?
2. How far apart are the subnet network addresses?

ADDRESS COUNT

The total number of addresses in a subnet is:

- $2^{\text{(host bits)}}$

Host bits are calculated as:

$32 - \text{prefix length}$

Example:

/27

$32 - 27 = 5$ host bits

$2^5 = 32$ total addresses

$32 - 2 = 30$ usable host addresses

BLOCK SIZE / INCREMENT

The block size, also called the subnet increment, tells you where each subnet begins.

For the octet in which subnetting occurs:

$256 - \text{subnet mask value} = \text{block size}$

Example:

/27 = 255.255.255.224

$256 - 224 = 32$

The subnet networks therefore increase by 32:

192.168.1.0

192.168.1.32

192.168.1.64

192.168.1.96

192.168.1.128

192.168.1.160

192.168.1.192

192.168.1.224

The address immediately before the next subnet begins is the broadcast address of the current subnet.

Example:

192.168.1.64/27

Next subnet: 192.168.1.96

Therefore:

Network address: 192.168.1.64
First usable host: 192.168.1.65
Last usable host: 192.168.1.94
Broadcast address: 192.168.1.95

PREFIX LENGTH AND SUBNET SIZE

Increasing the prefix length creates smaller networks with fewer available host addresses.

Examples:

/24 = 256 total addresses
/25 = 128 total addresses
/26 = 64 total addresses
/27 = 32 total addresses
/28 = 16 total addresses
/29 = 8 total addresses
/30 = 4 total addresses

Longer prefix = more network bits = fewer host bits = smaller subnet

Shorter prefix = fewer network bits = more host bits = larger subnet

WHAT SUBNETTING ACCOMPLISHES

Subnetting allows a larger IP network to be divided into smaller networks based on the needs of the network.

For example:

192.168.1.0/24

could be used as one network containing up to 254 usable host addresses.

Instead, it could be divided into smaller subnets for different:

Departments
VLANs
Physical locations
Network segments
Router-to-router links

Routers treat each subnet as a separate destination network.

Two hosts are considered to be on the same subnet when their IP addresses fall within the same network according to their subnet masks.

If two hosts are on different subnets, traffic between them normally needs to be routed by a Layer 3 device.

VLSM

VLSM stands for **Variable Length Subnet Mask**.

With VLSM, different subnets can use different prefix lengths depending on how many addresses each network requires.

Example:

192.168.1.0/25
192.168.1.128/26
192.168.1.192/27
192.168.1.224/28

This allows IPv4 address space to be allocated more efficiently because larger networks can receive larger subnets while smaller networks receive smaller subnets.

REMEMBER THESE FORMULAS IF NOTHING ELSE

****Usable host addresses – $2^{\text{hostbits}} - 2$ ****

****Number of subnets – $2^{\text{borrowedbits}}$ ****